polilabs: Knowledge-Representation Critique and the Passage-Dynamics Layer

June 9, 2026

Summary

This note responds to the latest feedback round: expand the corpus toward electorally consequential legislation (bills that move votes and donors, bipartisan "secret congress" laws, and failed bills with a record of *why* they failed), and critique/refine the knowledge representation so that agent queries are fast and the under/overcoverage problem shrinks. Three things were built: (1) a **passage-dynamics data layer** (roll-call votes, party splits, derived outcomes) covering all 232 corpus bills, fetched entirely from keyless public endpoints; (2) a new **`secret_congress` topic corpus** of 29 verified exemplar bills, from the CHIPS and Science Act to the failed Border Act of 2024; (3) three **aggregate query primitives** (`count_bills`, `get_bill_votes`, `find_bills_by_outcome`) plus a per-topic **facet-coverage contract** in `corpus_coverage()`. The critique below explains why these specific pieces, and what remains open.

1 What the representation could not say before tonight

The schema's strength has been *textual* fidelity: verbatim sections, canonical citations, local definitions, reified amendments. Its blind spot was *political* fidelity. Concretely, before this round the system could not answer any of the following, on any bill:

- Did this bill pass? (No outcome field anywhere.)
- Was the vote bipartisan? (No vote data at all; cosponsor party counts were the only proxy, and cosponsorship is a weak proxy — the Laken Riley Act had near-zero Democratic cosponsors and 12 Democratic Senate votes.)
- Why did this bill fail? (Action trails existed for only the ∼30% of AI bills that had Congress.gov enrichment; zero for redistricting.)
- How many bills are in the corpus from the 119th Congress? (The known eval failure: no count primitive, so agents paginated search results and counted — the dominant error source in the Opus-4.7-85%/Sonnet-4.6-70% gap.)

The meeting notes ask for exactly this layer: "how much it affected the vote," "show negative bills that failed (why it fails?)," "a lot of legislation is passed on a bipartisan vote in a single bill." These are not text questions; they are *passage-dynamics* questions, and the representation had no nouns for them.

2 The data layer: `billstatus.json`

One enrichment file per bill directory, written by `ingest/billstatus.py` and backfilled by `scripts/backfill_billstatus.py`. Three keyless sources: GovInfo BILLSTATUS bulk data (status, actions, structured sponsor/cosponsors, subjects, public-law numbers, recorded-vote pointers), House Clerk roll-call XML (per-party totals, per-member positions), Senate LIS roll-call XML (per-member positions, totals derived). No API key exists in the environment right now; none is needed for any of this.

Design decisions worth defending:

- **Derived outcome with evidence.** `outcome` is a 13-value enum (`enacted`, `failed_cloture`, `died_in_committee`, `reported_no_floor_vote`, `passed_house_only`, ...) derived from the action trail and roll-call results, never asserted bare: an `events` list carries each dated, verbatim action string behind the label. An agent asked "why did the Border Act fail?" quotes the cloture vote and its date instead of confabulating a narrative.
- **Bipartisanship as a number, labels second.** `bipartisan_support` = min over the two major parties of that party's yea-share on the vote, in $[0, 1]$. Labels (`party_line` $< 0.10 \leq$ `cross_party` $< 0.50 \leq$ `bipartisan` $< 0.90 \leq$ `near_unanimous`) are display sugar; the number is what filters and sorts. Calibration checks: CHIPS final House concurrence (24 R yeas) → 0.11; the COMPETES shell vote (1 R yea) → 0.005; PACT Act Senate → bipartisan.
- **Member positions only for outcome-determining votes.** Passage/cloture/veto-override votes carry all ∼435 member positions (party, state) — this is what enables delegation geography ("how did California vote on CHIPS?"). Amendment and procedural votes keep party totals only, bounding file sizes on amendment-heavy vehicles (NDAAs run 40+ roll calls).
- **The vehicle-bill trap is documented, not hidden.** A vehicle bill carries roll calls from an earlier life under another name (H.R. 4346's 2021 votes are America COMPETES votes), and conversely a bill's famous vote can live on a *different* bill number (KOSA's 91–3 Senate vote was on the S. 2073 vehicle, so S. 1409 itself correctly shows no floor vote — the curator note carries the explanation). The system prompt and tool descriptions instruct the agent to check vote dates/questions before attributing margins; `bills.bipartisan_support` uses the *final* passage-class vote for the same reason.

3 The corpus expansion: `secret_congress`

29 bills, hand-curated and *individually verified* against BILLSTATUS (title + public-law number) before inclusion — bill numbers were never trusted from memory; the one ambiguous case (DISCLOSE Act) was resolved to S. 4822 (117th) by checking the failed 49–49 cloture record. Six clusters defined in `corpus/secret_congress_criteria.md`:

Cluster	n	Exemplars
`quiet_bipartisan_law`	8	OSRA, postal reform, PACT, WRDA 2024, TAKE IT DOWN
`high_salience_bipartisan_law`	7	CHIPS, IIJA, Safer Communities, Laken Riley
`bipartisan_but_died`	7	Border Act, KOSA, Tax Relief Act, SAFE Banking, DISCLOSE
`absorbed_into_vehicle`	2	USICA/Endless Frontier, Electoral Count Reform Act
`omnibus_vehicle`	2	FY2023 omnibus, WRDA-2022→NDAA
`party_line_contrast`	3	IRA, OBBBA, Secure the Border Act

Notes on fidelity to the meeting notes: the "FRONTIERS Act" is the *Endless Frontier Act* (S. 1260, 117th), included with its real fate (passed the Senate 68–32 as USICA, died in conference, science core enacted inside CHIPS); the "water infrastructure bill" appears twice (S. 914, the 89–2 standalone, and H.R. 7776, WRDA-as-NDAA-vehicle); the donor angle is carried by the DISCLOSE Act (campaign-finance disclosure, failed cloture 49–49 on party lines) and by the antitrust pair (AICO, JCPA) that died without floor votes under platform lobbying. The single-bill bundling phenomenon is directly queryable: ECRA is `absorbed_into_vehicle` and H.R. 2617 is its `omnibus_vehicle`.

The clusters are curatorial; the vote record is mechanical. An agent that distrusts a cluster tag can recompute from `get_bill_votes`. This two-layer design (curated interpretation over mechanical evidence) follows the schema's existing derivation-typing principle for edges.

4 Query speed: aggregate-first, count-closing

The eval's failure mode was never retrieval quality — it was *shape*: questions whose answer is a set or a number were being served by a paginated-search loop. Each loop iteration costs a round trip (~1–3 s of model latency) and risks page-boundary undercounting. The three new primitives close the shapes that remained open:

- `count_bills(group_by, topic, congress, outcome, ...)` — exact SQL aggregate, one call, ~10 ms. Directly closes eval Q6 ("how many bills from the 119th?").
- `find_bills_by_outcome(outcome, cluster, min/max bipartisan_support, ...)` — the set-valued passage-dynamics query ("which bipartisan bills failed?").
- `get_bill_votes(bill_id)` — one bill's full fate in one call: outcome, public law, events, votes with party splits.

These follow the precedent of `find_bills_defining` / `find_bills_amending`: *when the question shape is known, give the agent a primitive that returns the complete answer with a coverage note, so "N results" is the whole truth rather than page one of it.*

5 Under/overcoverage: coverage as a contract

The deeper refinement is in `corpus_coverage()`. Previously it reported corpus-level metadata (congresses, tiers, known gaps as prose). It now reports, per topic, a **facet-coverage table**: for each data facet (`actions`, `cosponsors`, `subjects`, `votes`, `outcome`, `embeddings`), how many of the topic's bills carry that facet, computed live from the index so it cannot drift.

The point is epistemic, not cosmetic. Under- and overcoverage errors are both *coverage-belief* errors: the agent either believes the corpus knows less than it does (abstains wrongly — undercoverage of the answer space) or believes it knows more (asserts from a facet that is only 30%-populated — overcoverage). A live facet table converts both into checkable propositions: "`votes` covers 25/29 secret_congress bills, 6/12 redistricting bills, and 3/191 ai_governance bills (most AI bills die in committee — no roll call exists); `outcome` covers 232/232; embeddings cover 0 bills in this branch's `-skip-embeddings` build (7,880/29,616 ai_governance sections were embedded on the main checkout before the pass was handed off)." The agent's honesty rules can then be mechanical: before aggregating over a facet, check the facet is total; when it is not, say which slice the answer rests on. This is the same move the schema made for definitions (gap nodes instead of silent misses), applied to the corpus itself.

Two genuine overcoverage risks now exist and are guarded: vehicle-bill votes (§2) and partial dense-embedding coverage (the dense leg of hybrid search silently favors embedded bills; FTS covers all bills, and the facet table now exposes the imbalance instead of hiding it).

6 Geography and temporal

Both feedback items are served by existing fields once votes exist: `vote_positions(state, party, position)` answers delegation questions and state-level vote maps for the policymaker-facing visual layer; BILLSTATUS `subjects` include state names where bills touch them; `introduced_date`, action dates, vote dates, and `congress` give the temporal spine. A dedicated geography aggregate ("votes by state delegation") would be a natural fourth primitive if the frontend wants it; the SQL is a two-line GROUP BY on tables that now exist.

7 What is deliberately not claimed, and what is next

- **Media salience is asserted, not measured.** The secret-congress thesis is about coverage volume; the corpus encodes salience only through my cluster curation. A salience facet (e.g., GDELT or newspaper-archive mention counts per bill) would make `quiet_bipartisan_law` falsifiable. Medium effort; high value for the thesis.
- **"How much it affected the vote" (electoral outcomes) is not yet linkable.** Connecting member positions to subsequent vote-share changes needs election-results data (MIT Election Lab) and a member-district crosswalk, plus real identification care — this is a research design, not an ingest script. Flagged for a design discussion before building.
- **Donor data (FEC) is out of scope tonight.** The donor angle is currently represented by donor-*related* legislation (DISCLOSE) rather than donor behavior data.
- **Embeddings remain partial** (deferred to Raj's box per the earlier handoff); the new topic adds ∼130k sections to that queue, dominated by the four mega-vehicles.
- **Senate roll-call coverage for pre-1989 style URLs, amendment-level Senate votes inside BILLSTATUS,** and a handful of unresolved vote refs are recorded per-bill in `unresolved_vote_refs` rather than dropped silently.

Verification snapshot

232 bills indexed (191 `ai_governance`, 12 `redistricting`, 29 `secret_congress`); 172,887 sections; 347 roll-call votes; 0 parse errors. Outcome distribution: 88 pending, 84 died in committee, 24 reported-no-floor-vote, 20 enacted, 6 failed cloture, 10 other. Spot checks: CHIPS → `enacted`, PL 117–167, final House concurrence 243–187 (`bipartisan_support` 0.114); Border Act → `failed_cloture`, 43–50 on the motion to proceed; DISCLOSE → 49–49 cloture, `party_line`; S. 914 → `died_after_passing_senate` with the 89–2 passage vote; the two "enacted" AI-governance bills are the FY2024/FY2025 NDAAs (Tier-B AI riders), which is correct. 20,505 member-level vote positions indexed.